A PROPER MORPHISM THAT IS AN ISOMORPHISM OVER A QUASI-COMPACT OPEN BECOMES AN ISOMORPHISM AFTER AN ADMISSIBLE BLOWUP

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Editorial note. Counted span is the author's own proof environment, source lines 8950-8959, immediately following the statement at 8936-8948.

Editorial note. This chapter (spaces-more-morphisms) has no entry in reference/public-items.jsonl, so the range check is vacuous for it; the cross-lane label scan is the substantive check.

Editorial note. Three chapter lemmas are cited by \ref as navigation pointers into the same chapter.

Allowed standard prerequisites (editorial).

- Flatness and finite presentation after an admissible blowup. chapter lemma lemma-flat-after-blowing-up, cited at proof line 8951. Allows replacing the morphism by one that is flat and of finite presentation, at the cost of a U-admissible blowup.
- A proper, flat morphism over a scheme-theoretically dense open is finite. chapter lemma lemma-proper-flat-finite-over-dense-open, cited at proof line 8955. Turns properness plus flatness plus density of U into finiteness of f.
- Zariski's main theorem style criterion. chapter lemma lemma-zariski, cited at proof line 8956. Turns finiteness plus the fact that f is an isomorphism over U into an open immersion.
- Existence of a U-admissible blowup making U scheme-theoretically dense. used at proof lines 8953-8954. Produces the blowup after which the density hypothesis holds.

Source reference. Original lemma 087E: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8861–8861. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0B4J: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8625–8625. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0879: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8649–8649. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Let $U \subset B$ be an open subspace. Assume*

(1) *B is quasi-compact and quasi-separated,*
(2) *U is quasi-compact,*
(3) *$f : X \to B$ is proper, and*
(4) *$f^{-1}(U) \to U$ is an isomorphism.*

Then there exists a U-admissible blowup $B' \to B$ such that the strict transform X' of X maps isomorphically to B'.

Proof. By Lemma 087E we may assume that $X \to B$ is flat and of finite presentation. After replacing B by a U-admissible blowup if necessary, we may assume that $U \subset B$ is scheme theoretically dense. Then f is finite by Lemma 0B4J and an open immersion by Lemma 0879. Hence f is an open immersion whose image is closed and contains the dense open U, whence f is an isomorphism. □